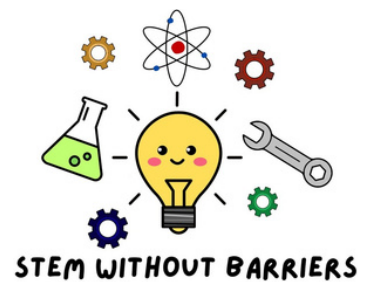


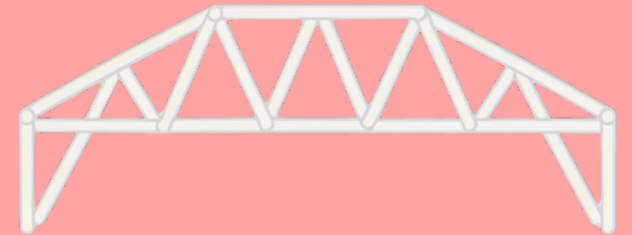
BRIDGE BUILDERS CHALLENGE

STEM ACTIVITY



MATERIALS:

- 20X PLASTIC STRAWS
- 1X ROLL OF MASKING TAPE
- 1X SCISSORS
- 35X WASHERS
- 1X PAPER INSTRUCTIONS
- 1X PAPER CUP



GOAL: USE PLASTIC STRAWS AND TAPE TO CREATE A BRIDGE, TRY YOUR BEST TO CARRY AS MUCH WASHERS AS YOU SHOULD

GATHER MATERIALS, AND MAKE SURE YOU HAVE ENOUGH SPACE. TAKE YOUR TIME AND PLAN OUT A DESIGN YOU'D LIKE! PAY ATTENTION TO SHAPES THAT WOULD HELP HOLD THE MOST WEIGHT.

TAPE THREE STRAWS TOGETHER TO MAKE A TRIANGLE SHAPE. MAKE SEVERAL OF THESE FOR THE BRIDGE.

LINE UP YOUR TRIANGLES AND TAPE THEM TOGETHER TO CREATE A LONG STRAIGHT "TRUSS" (WHICH IS A FANCY WORD FOR STRONG BRIDGE FRAME).

PLACE YOUR BRIDGE ENDS ON OBJECTS THAT ARE 12 INCHES APART (EX: BOOKS AT EQUAL HEIGHT ON EACH SIDE), THEN USE THE PAPER CUP, FILL IT WITH WASHERS TO COUNT HOW MUCH WEIGHT IT CAN HOLD

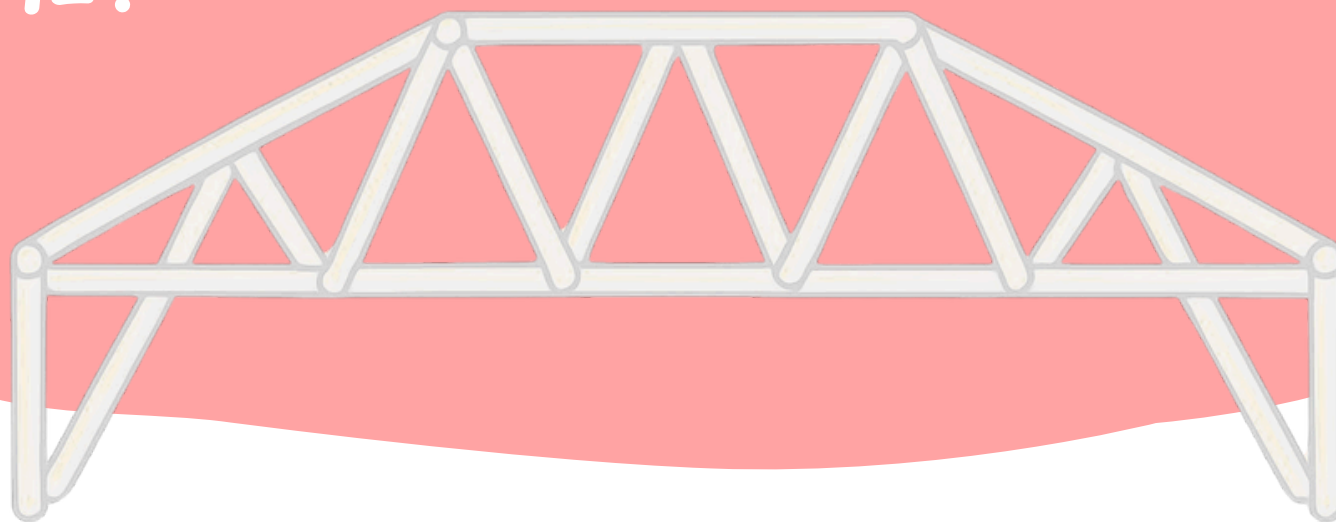
TAKE A PHOTO OF YOUR BRIDGE AND SHARE YOUR RESULTS WITH STEM WITHOUT BARRIERS IF YOU'D LIKE!



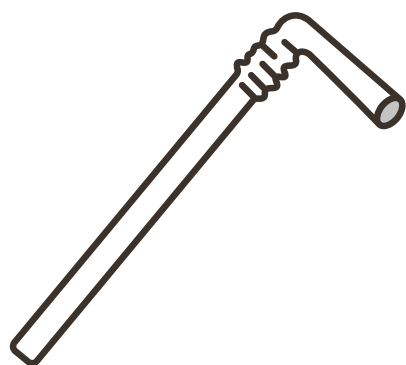
STEM CHALLENGE

BRIDGE BUILDERS CHALLENGE

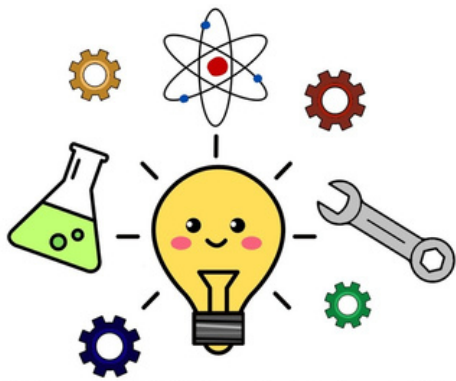
1. WHAT WAS THE STRONGEST PART OF YOUR BRIDGE?
2. WHAT WAS THE BIGGEST CHALLENGE?
3. WHAT WOULD YOU IMPROVE NEXT TIME?



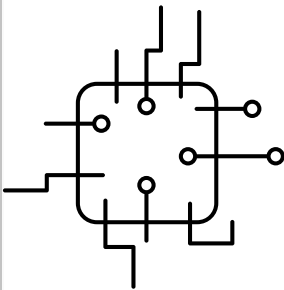
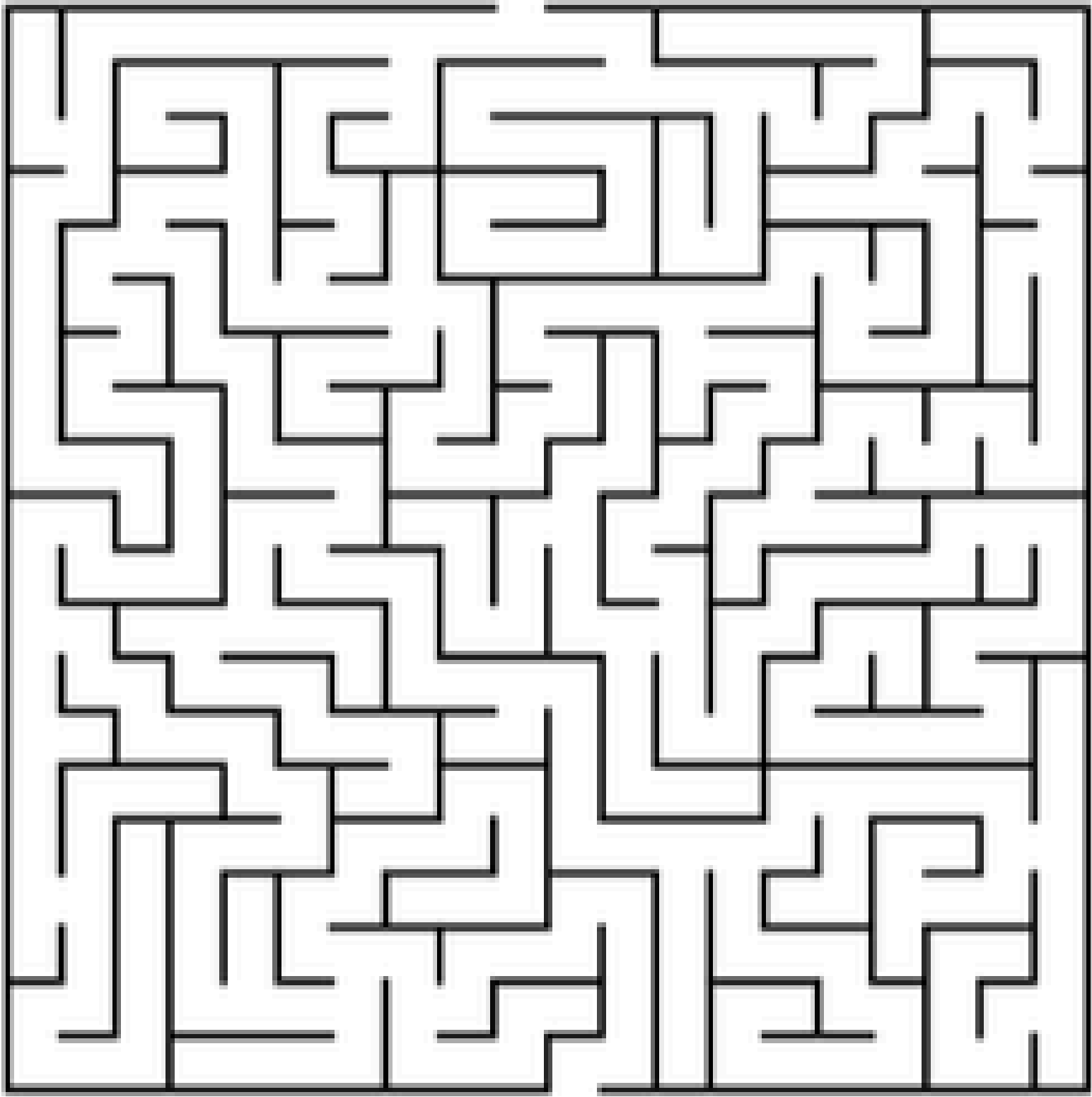
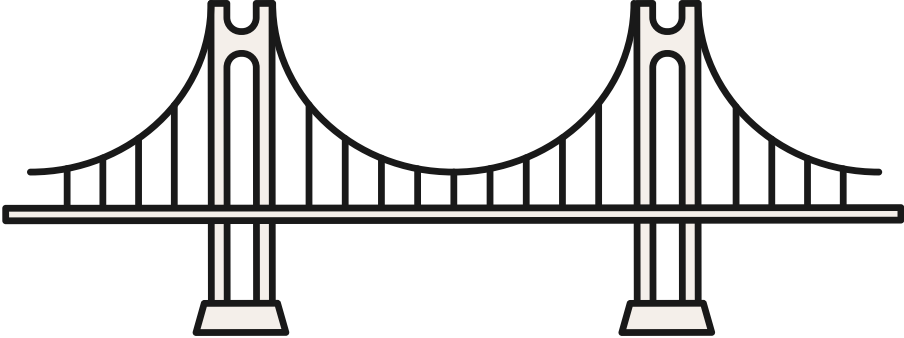
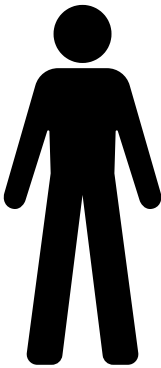
KEEP BUILDING. KEEP CREATING. KEEP
ENGINEERING.



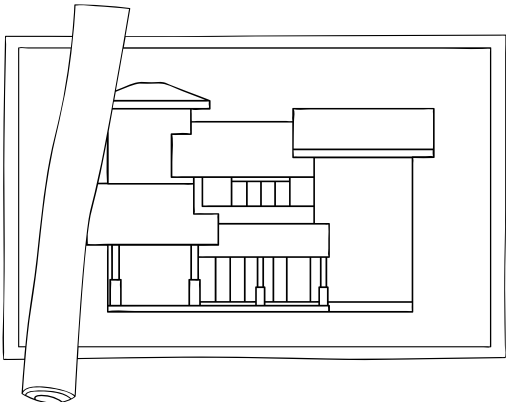
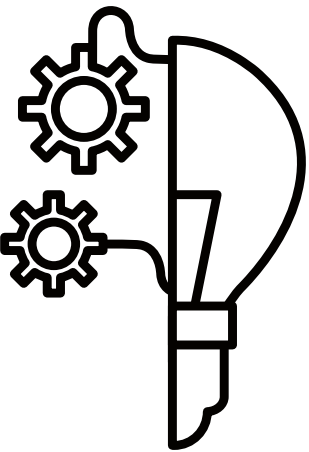
YOU ARE THE
FUTURE OF STEM!

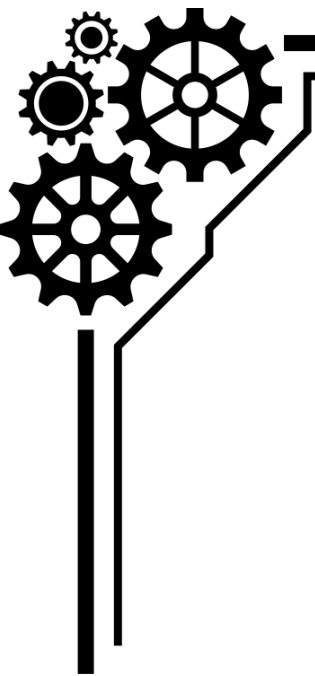


STEM WITHOUT BARRIERS

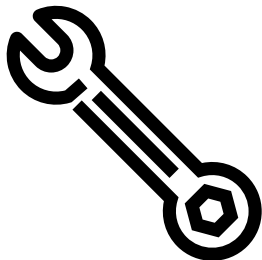


**HELP THE ENGINEER CROSS THE BRIDGE BY LEADING
THEM THROUGH THE BRIDGE!**

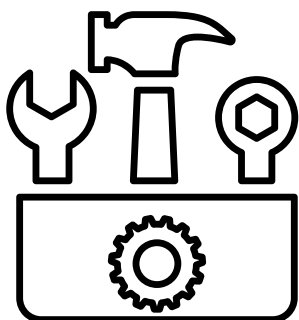
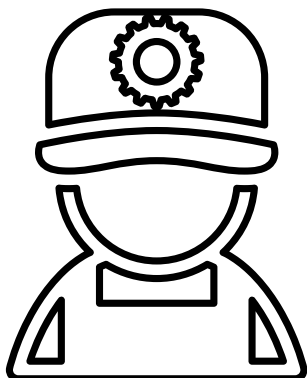
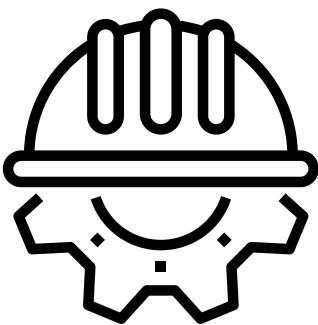




STEM WORDSEARCH



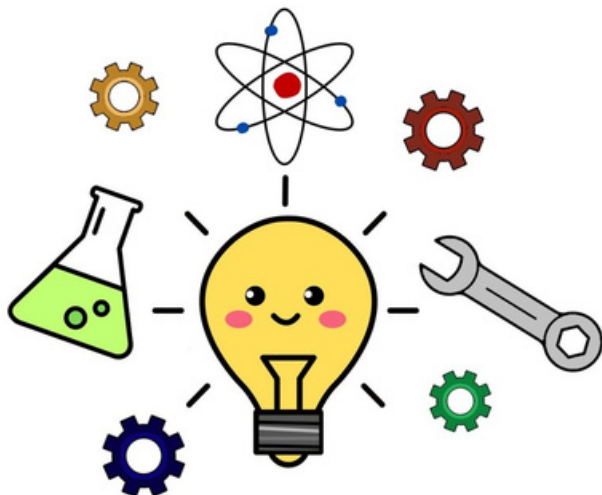
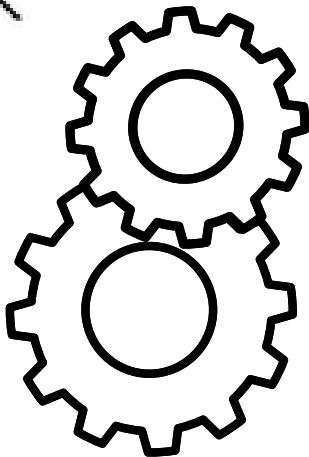
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K O V T B N M R F O U N D A T I O N D B
V Z I N G W E F K C U Y A B Q X H W M M
F A Z N N G A R V M N N R A B N C S I U
W F W Q T R S W O O A P C O K J E E X E
M Y E M P V U U S D F O H C F K N K Z S
J P I Y M M R W Q E W T I W L W G P N K
C M G P J A E O D L J E T S M A I J Y E
K B H I T X M Y E L Y C E T G I N R J T
D B T I W L E S S N G H C R X B E Y C C
M T H H U N N C I Z W N T E Z E E T S H
F P F N M I T A G X U O W N M A R R T P
B R I D G E C L N P B L L G X M S V R Y
T E B W J R G E P O U O Z T J Q U O U R
M A T E R I A L R F I G I H T R P G C E
J Q H R L X I K O L L Y R Y C H P M T X
J I G D F E N A C O D M A Y P H O O U B
N I S T R U S S E A P C N D E F R Q R A
P W S X L U V H S D R S F L U N T H E H
S P I W L B R S S K U E A N Y N U U F K



DESIGN
PROCESS
FOUNDATION
ARCHITECT
SKETCH
TRUSS
LOAD
MODEL

MEASUREMENT
STRUCTURE
MATERIAL
STRENGTH
JOINT
WEIGHT
BEAM

TECHNOLOGY
ENGINEER
BRIDGE
SUPPORT
SCALE
BUILD



STEM WITHOUT BARRIERS